Detecting Deepfakes with Metric Learning

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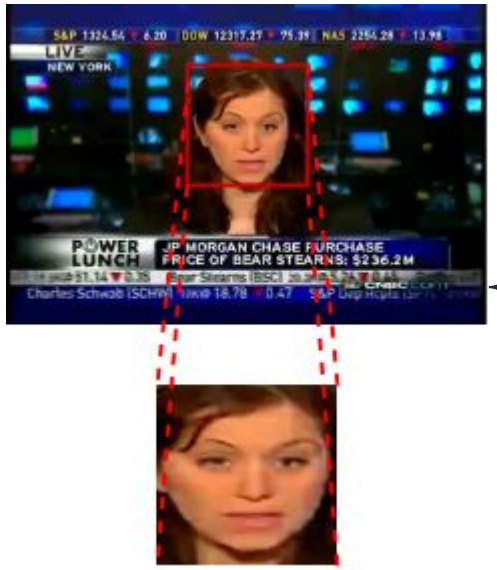


MTCNN
(Face Extraction)

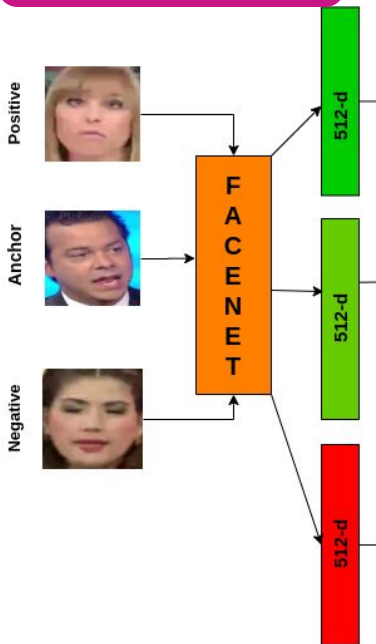
2D-CNN → Xception Net → Frame Classification

CNN+LSTM/GRU → Xception Net → Feature Extraction → LSTM

Sequence Classification

3D-CNN → Conv3D → Video Classification

Semi-Hard Triplets



Feature Classification

Model	Accuracy
Frames only	49.82
3d Convolution	43.47
Frames + LSTM/GRU	55.8
Triplets (Semi-hard)	**86.74**

TABLE II
FACEFORENSICS++(C40) ACCURACY PERFORMANCE

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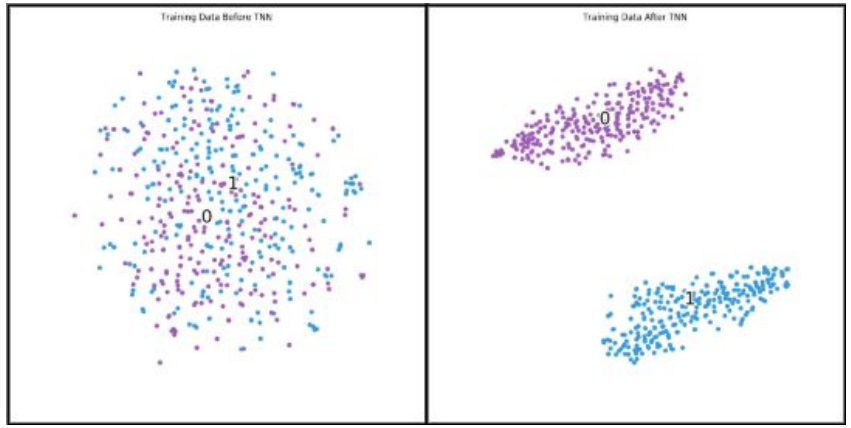
Face Forensics++



Model	P	R	F1	AUC	EER
Triplets + RF	89.84	82.73	86.14	92.85	12.14
Triplets + SGD	**90.55**	**82.74**	**86.47**	**92.9**	**10.07**

TABLE III

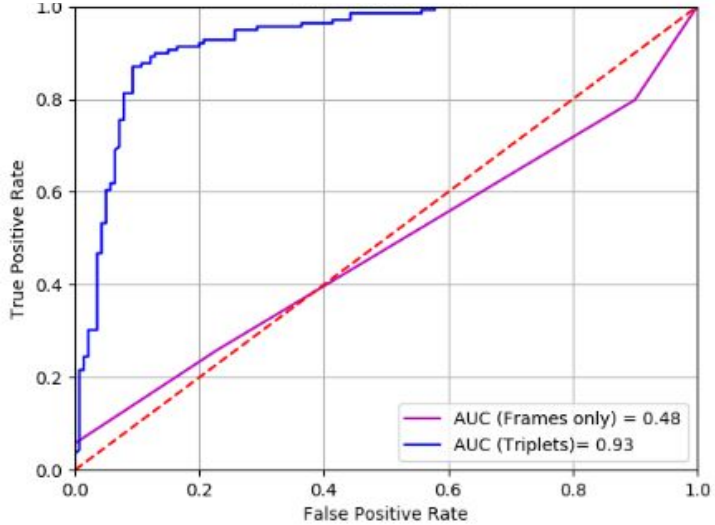
MODELS PERFORMANCE COMPARISON USING SEMI-HARD TRIPLETS. HERE, P=PRECISION, R=RECALL AND EER = EQUAL ERROR RATE.

Method	AUC Score
MesoInception4	53.6
Two-Stream	53.8
Muti-task	54.3
HeadPose	54.6
Meso-4	54.8
VA-MLP	55.0
VA-LogReg	55.1
FWA	56.9
Capsule	57.5
DSP-FWA	64.6
Xception	65.5
Ours	**99.2**

Model	NeuralTexture	Pristine
Steg. Features + SVM [14]	55.84	56.94
Cozzolino *et al.* [11]	62.15	56.27
Bayar and Stamm [10]	74.36	53.87
Rahmouni *et al.* [13]	59.99	56.79
MesoNet [9]	44.81	77.58
XceptionNet [12]	78.06	75.27
Ours	**90.71**	**82.73**

TABLE IV

COMPARISON OF OUR APPROACH TO PREVIOUS METHODS.



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